Javascript

Objects

Theory Notes

1. Creating Objects

- **Object literal** → simplest way.

```
let car = {};
```

- **Using** new Object() → creates an empty object.

```
let bike = new Object();
```

- **Constructor function** → reusable blueprint.

```
function Emp(id, name, salary){

   this.id = id;

   this.name = name;

   this.salary = salary;

}

let e = new Emp(101, "Ganesh", 40000);
```

2. Object Properties

- Access with **dot notation**: person.name.
- Access with **bracket notation**: person["bsc department"].
- Properties can be strings, numbers, or symbols.

3. Object Methods

- Object.assign(target, source) → merges objects.
- Spread operator {...obj1, ...obj2} → merges objects (ES6).
- Object.freeze(obj) → makes object immutable.
- Object.keys(obj) → returns array of property names.
- Object.values(obj) → returns array of property values.
- Object.entries(obj) → returns array of [key, value] pairs.

4. Important Concepts

- **Mutable vs Immutable** → objects are mutable; Object.freeze() prevents changes.
- **Nested objects** → objects can contain other objects.

- **Iteration** → use for...in or Object.entries() to loop through properties.

Practice Tasks

Short Answer

1. What is the difference between Object.assign() and the spread operator?
2. Why is Object.freeze() useful?
3. How do you access a property with spaces in its name?
4. What does Object.entries() return?

Coding Exercises

1. Create an object student with properties: id, name, course. Print them.
2. Write a constructor function Book with properties title, author, price. Create 2 book objects.
3. Merge two objects address and contact using both Object.assign and spread operator.
4. Freeze an object and try modifying its property. Observe the result.
5. Use Object.keys() to list all properties of an object.
6. Convert an object into a 2D array using Object.entries().
7. Iterate over an object using for...in and print keys and values.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Object</title>
</head>
<body>

    <h1>Object</h1>
    <!-- class Car{
        int speed;
        int mileage;

        void display(){
            sout(speed);
        }
    }

    Car obj = new Car(); -->
    <script>

        let car = {}; // object literal

        let bike = new Object(); //using new keyword

        console.log(typeof car);
```

```
console.log(typeof bike);

function emp(id, name, salary){
    this.id = id;
    this.name = name;
    this.salary = salary;
}

let e = new emp(103, "Ganesh", 40000);

console.log(e);
// constructor
console.log(typeof e);


let person ={
    name:"Arun",
    age:19,
    college:"karan",
   "bsc department" :"BSC"
}

let sports ={
    sport_name:"kabadi",
}

let mergedObject = Object.assign(person, sports);

let spreadOperator = {...person, ...sports};

console.log(spreadOperator);


console.log(mergedObject);

Object.freeze(person)
console.log(person);
console.log(person.age);
console.log(person["college"]);
console.log(person["bsc department"]);


person.name ="Kumari"


console.log(person.name);

console.log(Object.keys(person));
```

```
        let objectKeys = Object.keys(person);

        console.log(objectKeys[0]);

        console.log(Object.values(person));

        console.log(Object.entries(person));

        let twoDArray = Object.entries(person);

        console.log(twoDArray[0][0]);
    </script>
</body>
</html>
```